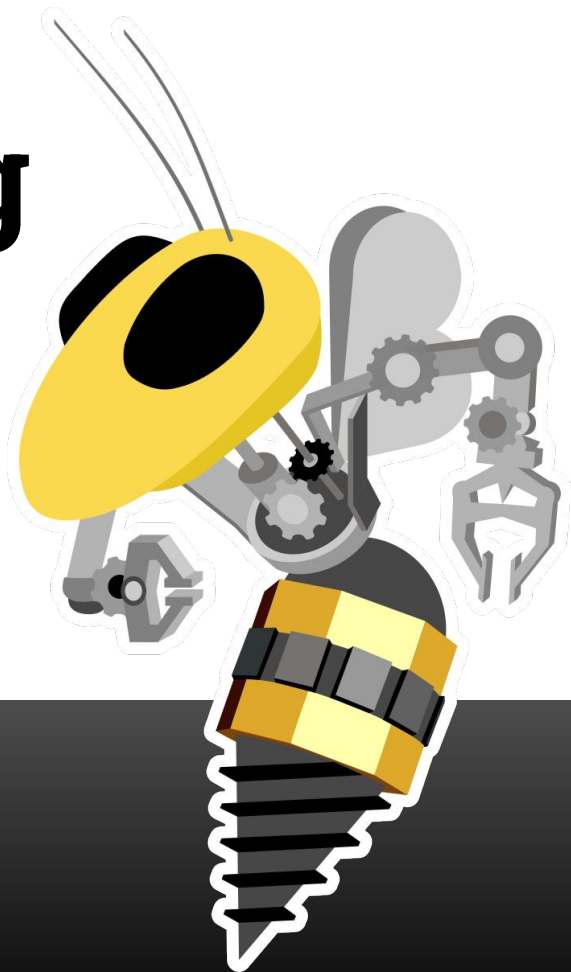


Firmware Training Week 4

Driver Design



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The screenshot shows the Arduino IDE Libraries Manager interface. The top bar includes tabs for 'main.cpp', 'PIO Home', and 'platformio.ini'. The left sidebar contains navigation icons for Home, Projects, Inspect, Libraries (selected), Boards, Platforms, and Devices. The main area is titled 'Registry' and includes a search bar with the text 'Search libraries...'. Below the search bar, there are tabs for 'Recently', 'Updated', 'Added', and 'Keywords'. The 'Recently' tab is active, showing a list of libraries with their names and update times. Below this, there are sections for 'Popular Tags' and 'Trending' libraries.

Registry | Installed | Built-in | Updates

Search libraries...

tft display dht* header-RH_ASK.h keyword:mqtt framework:mbed platform:espressif8266 more...

All Libraries Publish

Recently

Updated	Added	Keywords
Brasiliino about an hour ago	ChronoLog 4 hours ago	atmelavr
jescore about an hour ago	PedroRobot 9 hours ago	depth
ChronoLog 3 hours ago	SFM3X00 9 hours ago	encoder motor
BLE-Gamepad-Client 3 hours ago	ESPMeshMesh-dev 19 hours ago	esp32-idf
SHT4x 3 hours ago	HanumanMini 1 day ago	aircrack

Popular Tags

communication sensors control device display esp32 other data arduino

input i2c signal output esp8266 sensor processing timing lcd iot serial

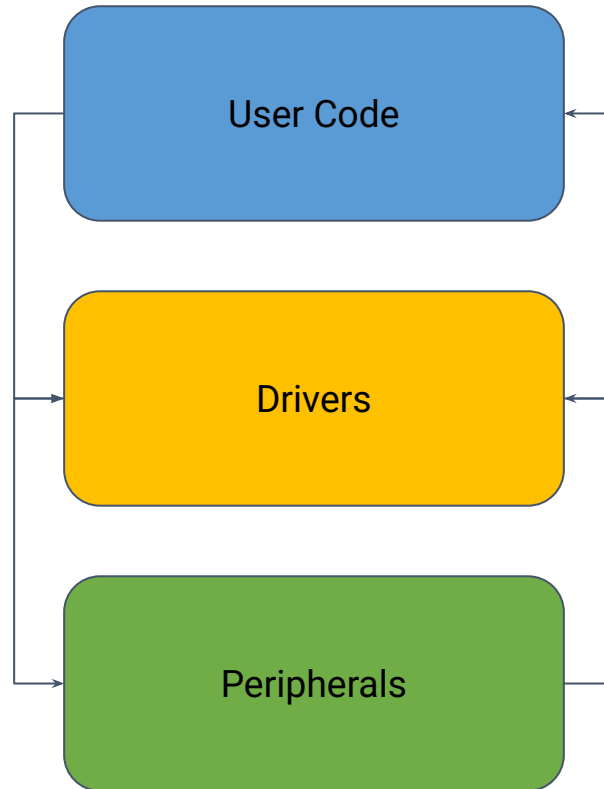
wifi tft spi temperature storage led oled graphics mqtt timer ethernet async http font

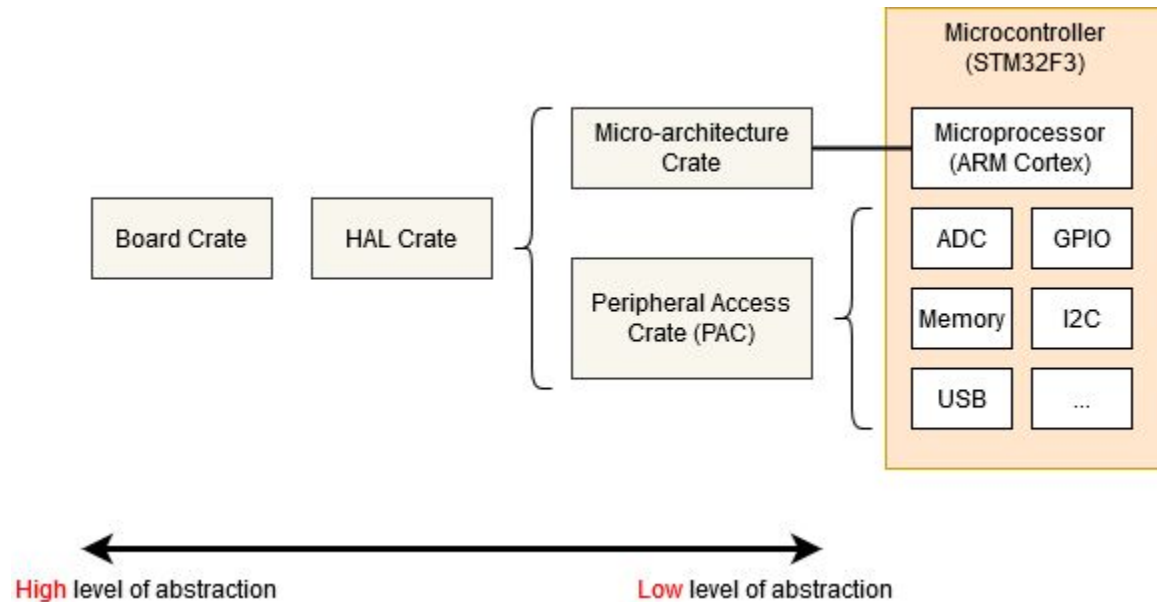
glcd pwm motor cariad driver stm32 library button webserver interrupt mbed time humidity eeprom

rtc server

Trending

Today	Week	Month
AutoConnect	Adafruit BusIO	Adafruit BusIO
EmblUI	ArduinoJson	Adafruit GFX Library
ArduinoWebsockets	Adafruit GFX Library	Adafruit TSL2561
RadioHead	Unity	Melopero RV3028
PageBuilder	Adafruit Unified Sensor	ArduinoJson
SimpleCollections	FastLED	Adafruit Unified Sensor
MD_Parola	Adafruit SSD1306	Unity





Macros

```
#define LED_PIN 13

uint8_t external_leds[] = {11, 10, 9, 3, 2};
size_t counter = 0;

void setup() {
    // Set pinMode of the onboard led so it can be used as an output
    pinMode(LED_PIN, OUTPUT);

    // Set the pinMode of each of the external leds so they can
    // be used as outputs
    for (size_t i = 0; i < 5; i++) {
        pinMode(external_leds[i], OUTPUT);
    }
}
```

```
#define SQUARE(x) (x * x)
#define PRINT(x) Serial.print(x);

void setup() {
    uint8_t value = SQUARE(2); // uint8_t value = 2 * 2;
    PRINT(value); // Serial.print(value);
}
```

```
#ifndef IMU_HPP
#define IMU_HPP

/**
 * ---CODE---
 */

#endif
```

```
#ifndef Arduino_h
uint8_t value = 32;
#else
uint8_t value = 100;
#endif
```

Enums

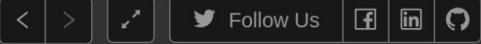
```
enum Direction {  
    NORTH = 0,  
    EAST = 1,  
    SOUTH = 2,  
    WEST = 3,  
};
```

```
enum ImuError {  
    // No error was detected during operation  
    NO_ERROR = 0,  
    // The IMU did not respond to the provided address  
    I2C_WRONG_ADDRESS = 1,  
    // The IMU was unable to be initialized properly  
    IMU_INIT_ERROR = 2,  
};
```

Principles of Driver Design

1. Know when to Reinvent the Wheel
2. Reference the Datasheet
3. Reference well-made drivers
4. Pretend you are the user

Know when to Reinvent the Wheel





Home



Projects



Inspect



Libraries



Boards



Platforms



Devices

RegistryInstalledBuilt-inUpdates

mpu6050

tft displaydht*header:RH_ASK.hkeyword:mqttframework:mbedplatform:espressif8266more...

Libraries58

Adafruit MPU6050 by Adafruit

82,4886Arduino

Arduino library for the MPU6050 sensors in the Adafruit shop

 [sensors](#)

 Infineon XMC, Kendryte K210, Atmel AVR, Atmel SAM, Espressif 8266, Intel ARC32, Microchip PIC32, Nordic nRF51, ST STM32, Teensy, TI MSP430, TI TIVA, Espressif 32, Nordic nRF52, ST STM8, Atmel megaAVR, Logic Green boards, Logicrom OpenCPU Development Platform, Raspberry Pi RP2040, K1921VK, TI MSP432, Heltec CubeCell, LibreTiny, Renesas RA

MPU6050 by Electronic Cats

21,4559Arduino

MPU6050 Arduino Library. MPU-6050 6-axis accelerometer/gyroscope Arduino Library.

 [sensors](#)

 [Atmel AVR](#), [Atmel SAM](#), [Espressif 8266](#), [ST STM32](#), [Espressif 32](#)

Reference the Datasheet

	MPU-6000/MPU-6050 Register Map and Descriptions	Document Number: RM-MPU-6000A-00 Revision: 4.2 Release Date: 09/19/2013
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CONTENTS

1	REVISION HISTORY	4
2	PURPOSE AND SCOPE	5
3	REGISTER MAP	6
4	REGISTER DESCRIPTIONS	9
4.1	REGISTERS 13 TO 16 – SELF TEST REGISTERS.....	9
4.2	REGISTER 25 – SAMPLE RATE DIVIDER	11
4.3	REGISTER 26 – CONFIGURATION.....	13
4.4	REGISTER 27 – GYROSCOPE CONFIGURATION.....	14
4.5	REGISTER 28 – ACCELEROMETER CONFIGURATION.....	15
4.6	REGISTER 35 – FIFO ENABLE.....	16
4.7	REGISTER 36 – I ² C MASTER CONTROL.....	17
4.8	REGISTERS 37 TO 39 – I ² C SLAVE 0 CONTROL.....	19
4.9	REGISTERS 40 TO 42 – I ² C SLAVE 1 CONTROL.....	22
4.10	REGISTERS 43 TO 45 – I ² C SLAVE 2 CONTROL.....	22
4.11	REGISTERS 46 TO 48 – I ² C SLAVE 3 CONTROL.....	22
4.12	REGISTERS 49 TO 53 – I ² C SLAVE 4 CONTROL.....	23
4.13	REGISTER 54 – I ² C MASTER STATUS.....	25
4.14	REGISTER 55 – INT PIN / BYPASS ENABLE CONFIGURATION.....	26
4.15	REGISTER 56 – INTERRUPT ENABLE.....	27
4.16	REGISTER 58 – INTERRUPT STATUS.....	28
4.17	REGISTERS 59 TO 64 – ACCELEROMETER MEASUREMENTS.....	29
4.18	REGISTERS 65 AND 66 – TEMPERATURE MEASUREMENT.....	30
4.19	REGISTERS 67 TO 72 – GYROSCOPE MEASUREMENTS.....	31
4.20	REGISTERS 73 TO 96 – EXTERNAL SENSOR DATA.....	32
4.21	REGISTER 99 – I ² C SLAVE 0 DATA OUT.....	34
4.22	REGISTER 100 – I ² C SLAVE 1 DATA OUT.....	34
4.23	REGISTER 101 – I ² C SLAVE 2 DATA OUT.....	35
4.24	REGISTER 102 – I ² C SLAVE 3 DATA OUT.....	35
4.25	REGISTER 103 – I ² C MASTER DELAY CONTROL.....	36
4.26	REGISTER 104 – SIGNAL PATH RESET.....	37
4.27	REGISTER 106 – USER CONTROL.....	38
4.28	REGISTER 107 – POWER MANAGEMENT 1.....	40
4.29	REGISTER 108 – POWER MANAGEMENT 2.....	42

	MPU-6000/MPU-6050 Register Map and Descriptions	Document Number: RM-MPU-6000A-00 Revision: 4.2 Release Date: 09/19/2013
---	--	---

4.30	REGISTER 114 AND 115 – FIFO COUNT REGISTERS.....	43
4.31	REGISTER 116 – FIFO READ WRITE.....	44
4.32	REGISTER 117 – WHO AM I.....	45

Reference well-made
drivers



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RobTillaart

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Arduino library maker, Software Engineer, worked at different research labs in high tech.

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[Gemert](#) Netherlands

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Sponsors



Achievements



Pretend you are the
user

```
/**  
 * Read the value stored in a given register  
 */  
uint8_t read_register(uint8_t regno);
```

VS

```
/**  
 * Read the x-axis reading of the gyroscope  
 */  
int16_t read_gyro_x();
```

Lab